Hybrid Classical–Neural Detection of Voice Pathology from Sustained Vowels with Cross-Database Evaluation

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***Abstract*—We propose a non-invasive screening system that labels a short sustained-vowel recording as *healthy* or *pathological*. The system combines hand-implemented classical speech analysis—the short-time Fourier transform (STFT), mel-frequency cepstral coefficients (MFCC), linear predictive coding (LPC), and pitch-perturbation measures—with a lightweight convolutional neural network (CNN) in a late-fusion architecture. An MFCC–Gaussian mixture model (GMM) classifier serves as the baseline. The system is trained on the public Saarbrücken Voice Database (SVD) under a speaker-independent protocol and tested, without retraining, on the independent VOICED database recorded with different equipment.**

***Index Terms*—voice pathology detection, MFCC, linear prediction, jitter and shimmer, CNN, cross-database evaluation**

I. Problem and Motivation

Pathologies such as nodules, polyps, laryngitis, and vocal-fold paralysis change how the vocal folds vibrate. The result is a voice that sounds hoarse, breathy, or rough. A clinical examination is the reference standard, but it requires an otolaryngologist and dedicated equipment. A recording-based screening tool could flag at-risk speakers early and support remote follow-up. It is a decision-support aid, not a replacement for diagnosis.

Many studies report high accuracy on the SVD. However, Huckvale and Buciuleac [1] showed that these numbers are hard to compare: papers use different pathology subsets, audio materials, and cross-validation schemes. When three published systems were re-implemented under one protocol, the differences between them largely disappeared. End-to-end deep learning on small clinical data is not automatically better either: a convolutional–recurrent network trained on raw SVD vowels reached 68% accuracy [2].

In the source–filter model of speech production [3], voice pathologies act mainly on the *source* (vocal-fold vibration). Spectral-envelope features such as MFCCs [4] mostly describe the *filter* (the vocal tract). We therefore combine envelope features, source-related measures (LPC residual, F0 perturbation), and a learned spectro-temporal representation, and evaluate them under a strict protocol.

II. Research Questions

RQ1: Under a speaker-independent protocol, does late fusion of a CNN on log-mel spectrograms with interpretable MFCC/LPC/perturbation features outperform an MFCC–GMM baseline? **RQ2:** Which feature family—envelope (MFCC), source (LPC residual, jitter, shimmer), or learned (CNN)—transfers best to a database recorded with different equipment and at a different sampling rate? **RQ3:** Is performance consistent across sexes, or does the classifier rely on sex-related cues?

III. Data

SVD [5], [6]: German speakers recorded at 50 kHz, 16-bit. It contains 1002 speakers with a wide range of voice disorders and 851 controls [1]. Each session has /a/, /i/, /u/ at normal, high, low, and rising–falling pitch, plus one sentence. The database is released on Zenodo under CC BY 4.0 (DOI 10.5281/zenodo.16874898). Of the 38 GB release, only the normal-pitch /a/ recordings are used, converted from `.nsp` to `.wav` with the published converter.

VOICED [7]: 208 recordings of a 5 s sustained /a/, sampled at 8 kHz and released on PhysioNet under the ODC Attribution License. It is used *only* as an external test set.

TABLE I
Databases (SVD: speakers in the full database [1]).

Database	Healthy / Path.	Fs	Role
SVD	851 / 1002 spk.	50 kHz	train / CV
VOICED	58 / 150 rec.	8 kHz	external test

Task: binary classification, healthy vs. any pathology. Sessions from the same speaker are always kept in the same fold to prevent speaker leakage (some SVD speakers have many sessions [1]). For the cross-database experiment, the SVD is downsampled to 8 kHz with an anti-aliasing filter so that the training and test bandwidths match.

IV. Methodology

Front end (implemented from scratch in NumPy, CS 568 topics):

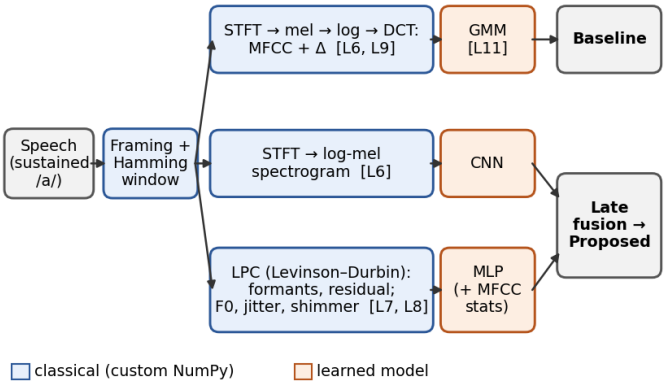


Fig. 1. Pipeline. Blue: classical analysis implemented from scratch (lecture numbers in brackets). Orange: learned models.

- *STFT and MFCC* (L6, L9): pre-emphasis, 25 ms Hamming-windowed frames with a 10 ms hop, and an FFT, followed by a 40-band mel filterbank, logarithm, and DCT-II. This gives 13 MFCCs plus Δ coefficients; the same filterbank output without the DCT is the log-mel spectrogram used by the CNN.
- *LPC* (L7): order-16 LPC with the autocorrelation method and the Levinson–Durbin recursion [8]. From it, the first three formants and their bandwidths (polynomial roots) are derived. Inverse filtering gives the prediction residual $e[n] = s[n] - \sum_{k=1}^{p} a_k\, s[n-k]$, an estimate of the excitation. Its normalized energy and kurtosis are computed.
- *Source/perturbation* (L8): F0 is estimated by autocorrelation. Local jitter and shimmer are computed from consecutive period lengths T_i and peak amplitudes A_i; for example, jitter $= \frac{1}{N-1} \sum_i |T_i - T_{i+1}| \,/\, \bar{T}$.

Frame-level features are summarized per recording (mean, standard deviation); every block is unit-tested against a reference library.

Hypothesis: pathologies disturb the regularity of the glottal source more than the vocal-tract envelope. Source descriptors should therefore add information that MFCCs and spectrograms do not capture, especially when the recording channel changes.

Baseline: two diagonal-covariance GMMs [9] (healthy, pathological) trained on frame-level MFCC+Δ vectors. A recording is classified by the log-likelihood ratio summed over its frames, and the number of mixtures is chosen by inner cross-validation.

Proposed (Fig. 1):

- a small CNN (three convolutional blocks with batch normalization and pooling, then global average pooling) on a fixed-length log-mel excerpt;
- a two-layer MLP on the standardized vector of MFCC statistics, LPC features, and perturbation measures;
- concatenation of the two embeddings followed by a sigmoid output (late fusion).

Training uses class-weighted binary cross-entropy and early stopping. **Ablations:** CNN only, envelope features only, source features only, and full fusion.

V. Evaluation Protocols

P1: speaker-independent, stratified 5-fold cross-validation on the SVD, following the common protocol of [1]. **P2:** cross-database transfer—train on the full SVD at 8 kHz, test on VOICED without fine-tuning (RQ2). **P3:** results broken down by sex, plus a sex-balanced re-sampled training run (RQ3). **P4:** the ablations above, evaluated under both P1 and P2. **Metrics:** unweighted average recall (UAR, the primary metric because the classes are imbalanced), F1, sensitivity, specificity, and ROC AUC, reported as mean ± standard deviation over folds. Systems are compared with paired tests over folds.

Compute and reproducibility: Google Colab with data, cached `.npy` features, and checkpoints on Google Drive. Code, fold definitions, and configurations will be public on GitHub (https://github.com/mmerts/voice-pathology-detection).

Risks and mitigations: (i) small, imbalanced data—a compact CNN, class weighting, and early stopping; (ii) pitch tracking may fail on severely pathological voices—frames where tracking fails are masked, and their proportion is itself used as a feature; (iii) the sampling-rate mismatch between the databases—resolved by training at 8 kHz for P2.

Timeline: by 20 Nov, data and front end with unit tests; by 4 Dec, GMM baseline and the cross-validation framework; by 18 Dec, CNN, fusion, ablations, and P2–P3; by 25 Dec, final paper and code; 28 Dec–1 Jan, presentation.

Expected outcomes: a reproducible, speaker-independent comparison of classical, learned, and fused features; an estimate of how much accuracy survives a change of database; and evidence on whether source descriptors add information beyond envelope and CNN features.

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